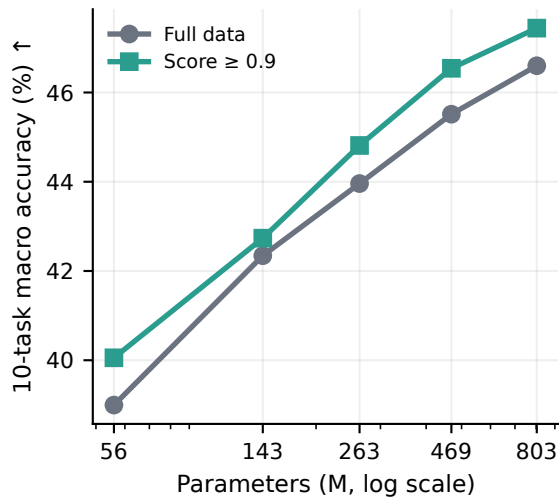
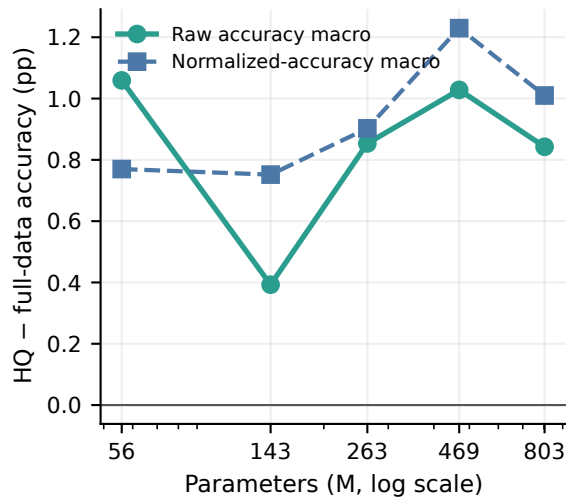


Exploratory data-quality scale sweep: fixed 20B-token budget

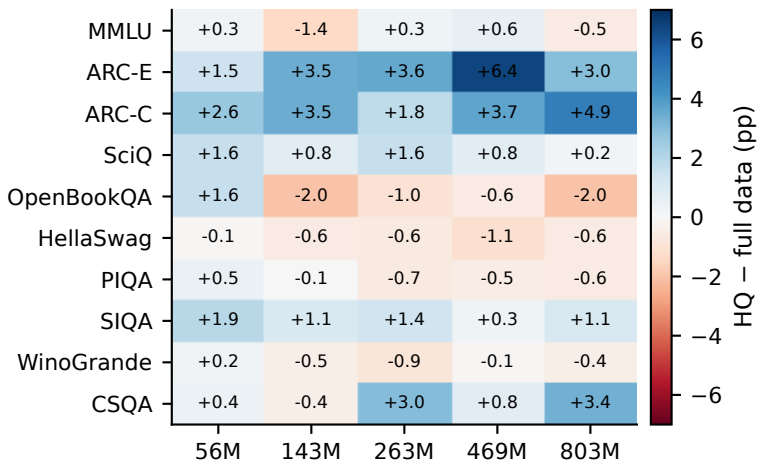
(a) Final score at 20B tokens



(b) Final quality-filtering gain



(c) Task-level gap at 20B tokens (pp)



(d) Gain along each single training run

